Statistical Machine Learning - Fundamentals - Classification Learning - 1

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Chunk

```
## Warning: package 'kernlab' was built under R version 4.5.2

## Warning: package 'mlbench' was built under R version 4.5.2

## Warning: package 'bestglm' was built under R version 4.5.2

## Loading required package: leaps

## Warning: package 'leaps' was built under R version 4.5.2

## Warning: package 'glmnet' was built under R version 4.5.2

## Loading required package: Matrix

## Loaded glmnet 4.1-10

## Warning: package 'ggplot2' was built under R version 4.5.2

##
## Attaching package: 'ggplot2'

## The following object is masked from 'package:kernlab':
##
##     alpha

## Warning: package 'BMS' was built under R version 4.5.2

## corrplot 0.95 loaded

## Loading required package: carData

## Warning: package 'ROCR' was built under R version 4.5.2
```

Introduction to the problem and the data

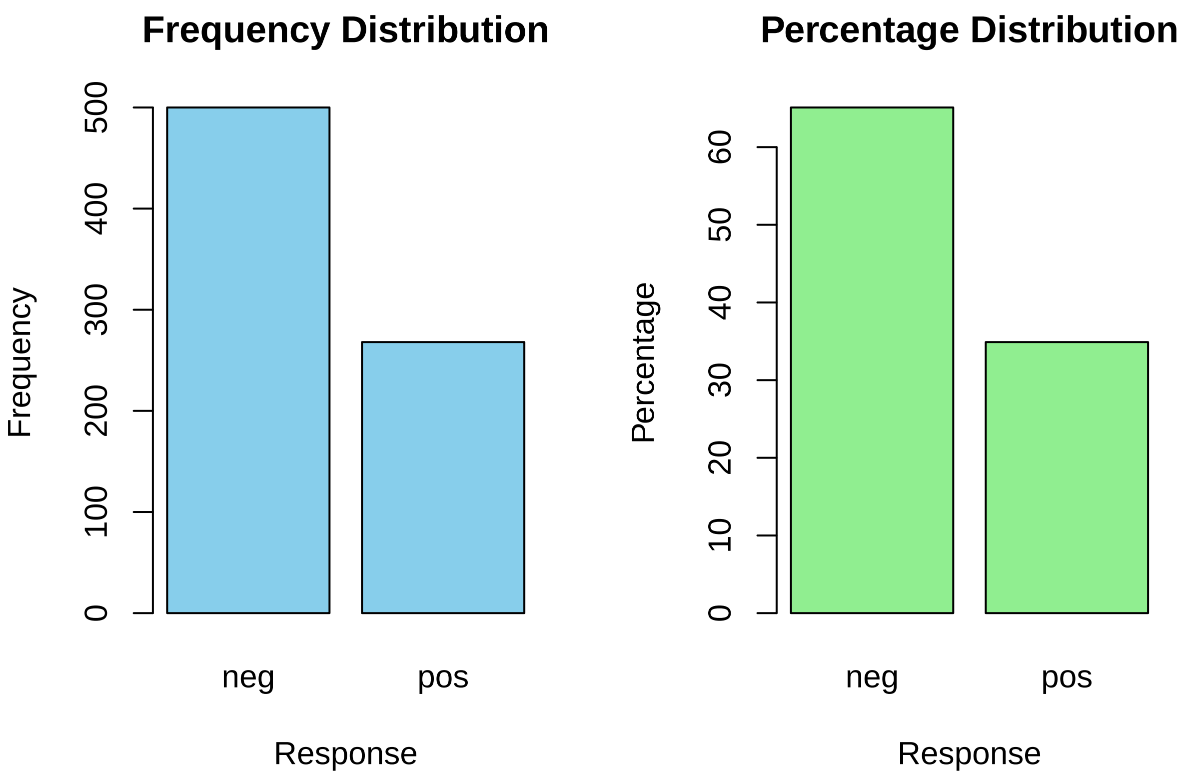
We are first studying the famous Pima Indian Diabetes data originating from the area of Phoenix in Arizona. This data featuring 768 females of Pima Indian ancestry for which it become important to not only classify but also comprehend the risk factors for diabetes!

```
##   pregnant glucose pressure triceps insulin mass pedigree age diabetes
## 1        6     148       72      35       0 33.6    0.627  50      pos
## 2        1      85       66      29       0 26.6    0.351  31      neg
## 3        8     183       64       0       0 23.3    0.672  32      pos
## 4        1      89       66      23      94 28.1    0.167  21      neg
## 5        0     137       40      35     168 43.1    2.288  33      pos
## 6        5     116       74       0       0 25.6    0.201  30      neg

##     pregnant glucose pressure triceps insulin mass pedigree age diabetes
## 763        9      89       62       0       0 22.5    0.142  33      neg
## 764       10     101       76      48     180 32.9    0.171  63      neg
## 765        2     122       70      27       0 36.8    0.340  27      neg
## 766        5     121       72      23     112 26.2    0.245  30      neg
## 767        1     126       60       0       0 30.1    0.349  47      pos
## 768        1      93       70      31       0 30.4    0.315  23      neg

## Sample size (n): 768

## Dimensionality of input space (p): 8

## Position of the response variable (py): 9

## Number of missing values: 0

## Richness status: richness

##   pregnant glucose pressure triceps insulin mass pedigree age   Y
## 1        6     148       72      35       0 33.6    0.627  50 pos
## 2        1      85       66      29       0 26.6    0.351  31 neg
## 3        8     183       64       0       0 23.3    0.672  32 pos
## 4        1      89       66      23      94 28.1    0.167  21 neg
## 5        0     137       40      35     168 43.1    2.288  33 pos
## 6        5     116       74       0       0 25.6    0.201  30 neg

##     pregnant glucose pressure triceps insulin mass pedigree age   Y
## 763        9      89       62       0       0 22.5    0.142  33 neg
## 764       10     101       76      48     180 32.9    0.171  63 neg
## 765        2     122       70      27       0 36.8    0.340  27 neg
## 766        5     121       72      23     112 26.2    0.245  30 neg
## 767        1     126       60       0       0 30.1    0.349  47 pos
## 768        1      93       70      31       0 30.4    0.315  23 neg
```

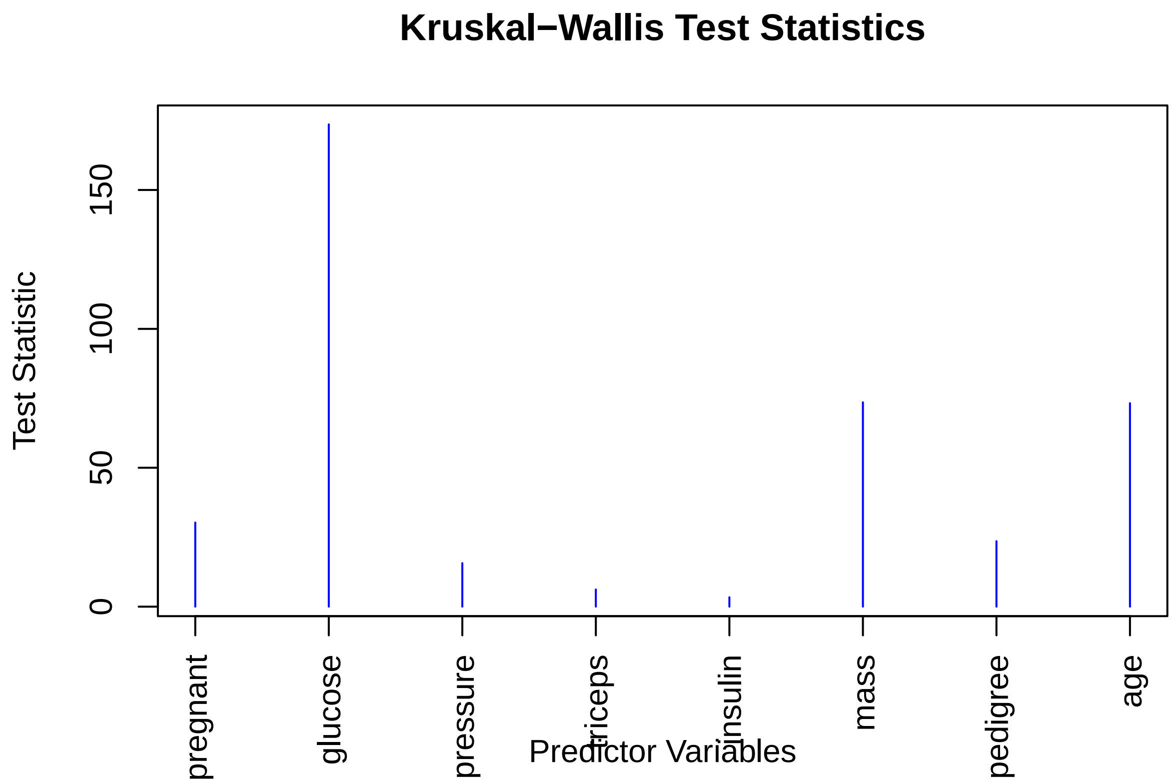
Exploration of the Response Variable

Let's now explore the distribution of the response variable along with the idiosyncratic assessment of the individual discriminative and predictive power of each predictor (explanatory) variable.

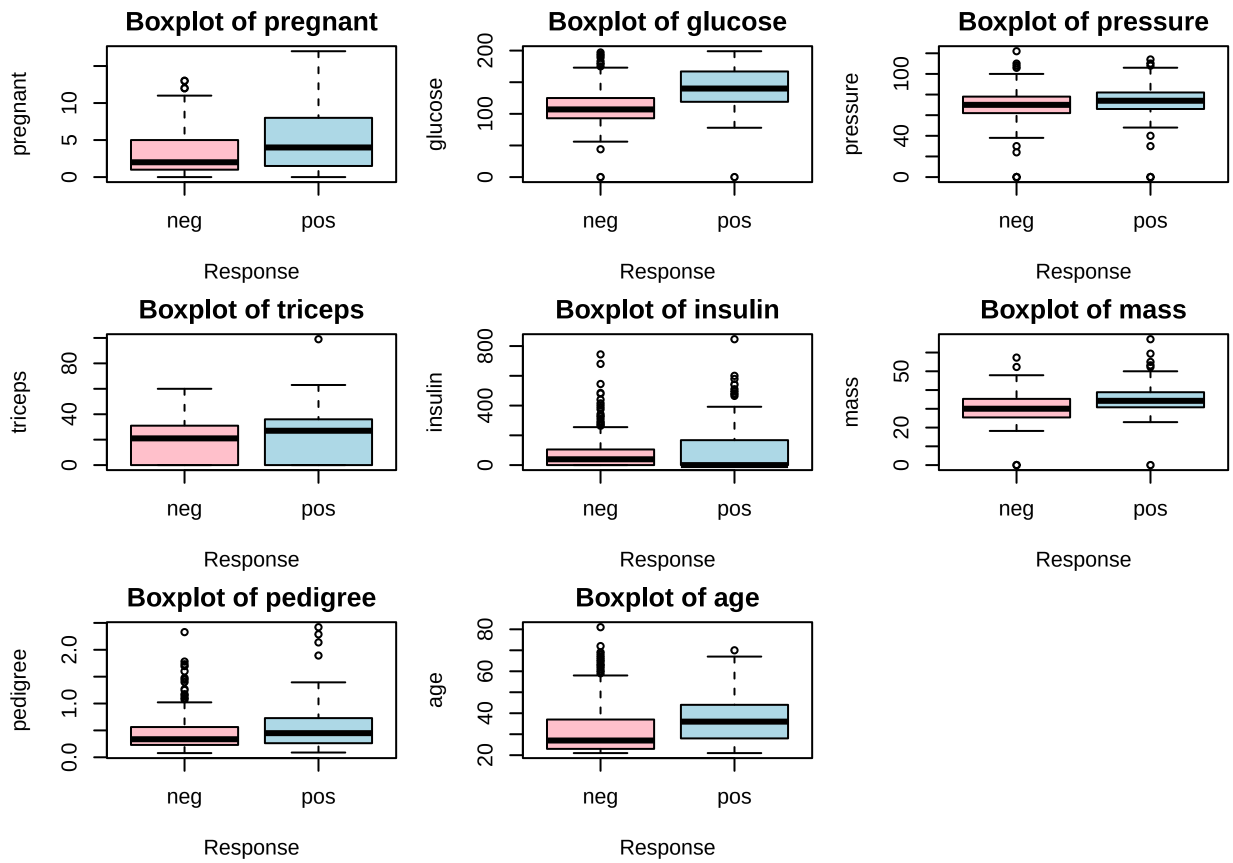
```
## Frequency distribution:

##
## neg pos
## 500 268

##
## Percentage distribution:

##
##      neg      pos
## 65.10417 34.89583
```

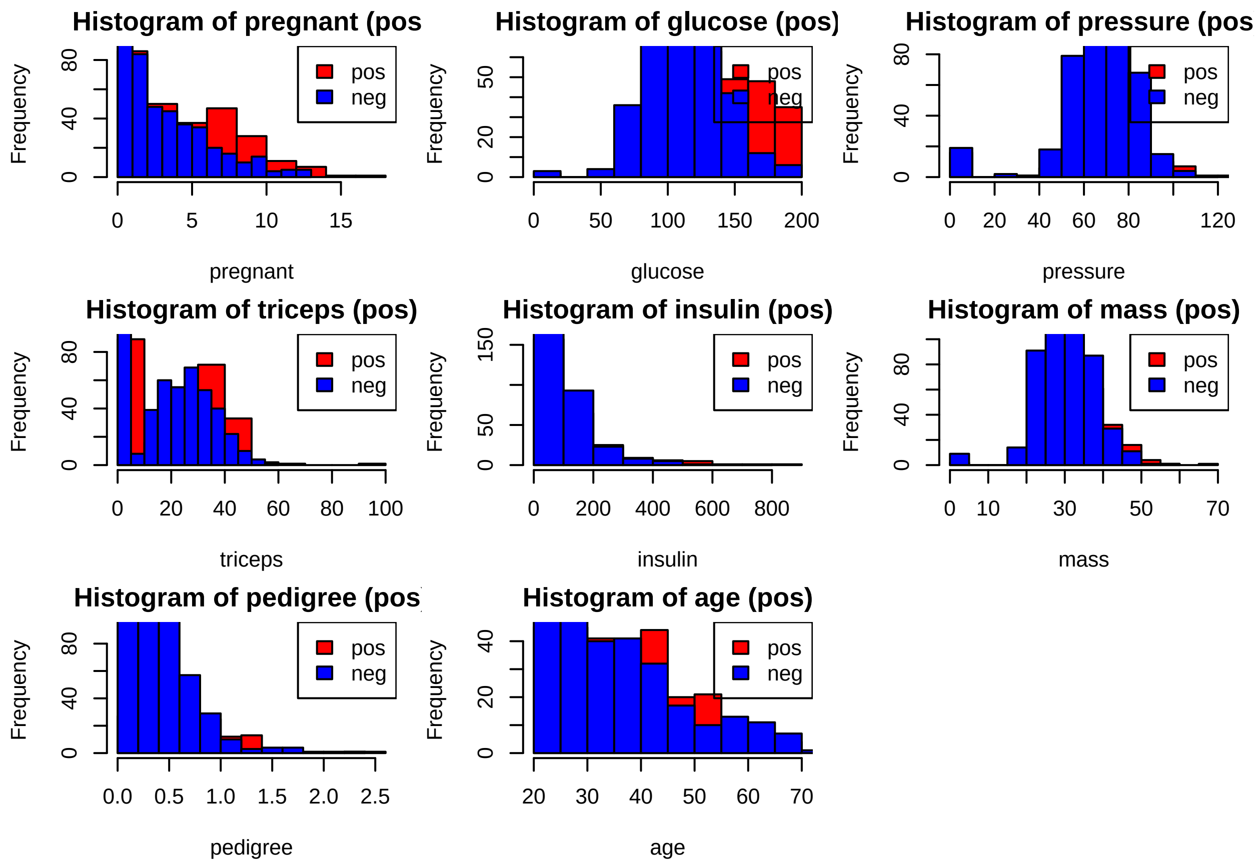



```
## Max/Min Ratio: 1.865672

## Imbalance Status: reasonable balance
```



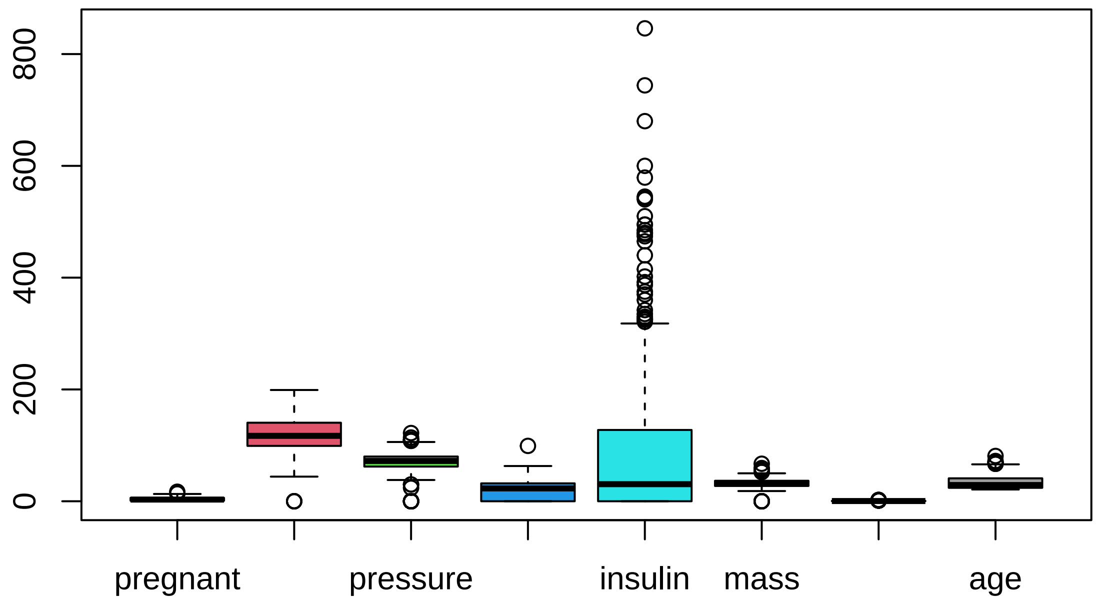

```
## Best Predictor: glucose

## Worst Predictor: insulin
```

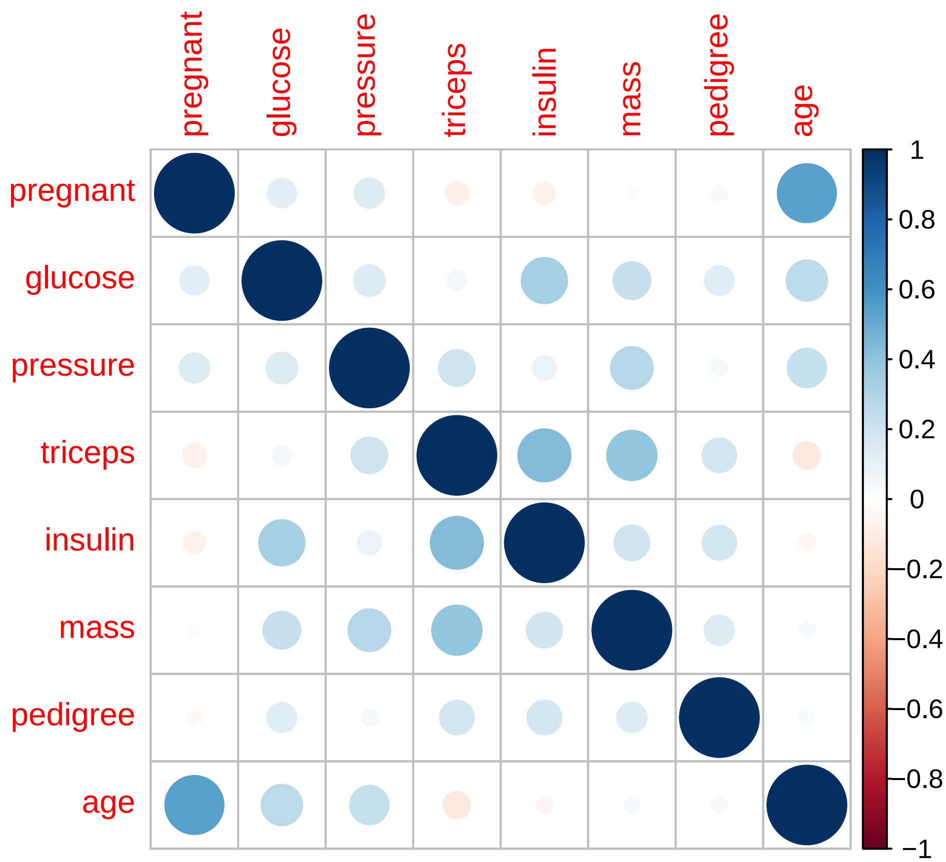

Boxplot of pregnant
Boxplot of glucose
Boxplot of pressure
Boxplot of triceps
Boxplot of insulin
Boxplot of mass
Boxplot of pedigree
Boxplot of age
pregnant
glucose
pressure
triceps
insulin
mass
pedigree
age
neg
pos
Response



Correlation analysis and input in-depth look

```
## 'data.frame':    768 obs. of  8 variables:
##  $ pregnant: num  6 1 8 1 0 5 3 10 2 8 ...
##  $ glucose : num  148 85 183 89 137 116 78 115 197 125 ...
##  $ pressure: num  72 66 64 66 40 74 50 0 70 96 ...
##  $ triceps : num  35 29 0 23 35 0 32 0 45 0 ...
##  $ insulin : num  0 0 0 94 168 0 88 0 543 0 ...
##  $ mass    : num  33.6 26.6 23.3 28.1 43.1 25.6 31 35.3 30.5 0 ...
##  $ pedigree: num  0.627 0.351 0.672 0.167 2.288 ...
##  $ age     : num  50 31 32 21 33 30 26 29 53 54 ...
```


800
600
400
200
0
pregnant
pressure
insulin
mass
age



Introduction to Fitting logistic Regression

```
##
## Call:
## glm(formula = Y ~ ., family = binomial(link = "logit"), data = XY)
##
## Coefficients:
##                Estimate Std. Error z value Pr(>|z|)
## (Intercept) -8.4046964  0.7166359 -11.728  < 2e-16 ***
## pregnant     0.1231823  0.0320776   3.840 0.000123 ***
## glucose      0.0351637  0.0037087   9.481  < 2e-16 ***
## pressure    -0.0132955  0.0052336  -2.540 0.011072 *
## triceps      0.0006190  0.0068994   0.090 0.928515
## insulin     -0.0011917  0.0009012  -1.322 0.186065
## mass         0.0897010  0.0150876   5.945 2.76e-09 ***
## pedigree     0.9451797  0.2991475   3.160 0.001580 **
## age          0.0148690  0.0093348   1.593 0.111192
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for binomial family taken to be 1)
##
##     Null deviance: 993.48  on 767  degrees of freedom
## Residual deviance: 723.45  on 759  degrees of freedom
## AIC: 741.45
```

```
##
## Number of Fisher Scoring iterations: 5
```

Chunk

```
##
## Call:
## glm(formula = diabetes ~ ., family = binomial(link = "logit"),
##     data = PimaIndiansDiabetes)
##
## Coefficients:
##              Estimate Std. Error z value Pr(>|z|)
## (Intercept) -8.4046964  0.7166359 -11.728  < 2e-16 ***
## pregnant     0.1231823  0.0320776   3.840 0.000123 ***
## glucose      0.0351637  0.0037087   9.481  < 2e-16 ***
## pressure    -0.0132955  0.0052336  -2.540 0.011072 *
## triceps      0.0006190  0.0068994   0.090 0.928515
## insulin     -0.0011917  0.0009012  -1.322 0.186065
## mass         0.0897010  0.0150876   5.945 2.76e-09 ***
## pedigree     0.9451797  0.2991475   3.160 0.001580 **
## age          0.0148690  0.0093348   1.593 0.111192
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for binomial family taken to be 1)
##
##     Null deviance: 993.48  on 767  degrees of freedom
## Residual deviance: 723.45  on 759  degrees of freedom
## AIC: 741.45
##
## Number of Fisher Scoring iterations: 5
```

```
## Chi-Squared Statistic: 270.0385
```

```
## Degrees of Freedom: 8
```

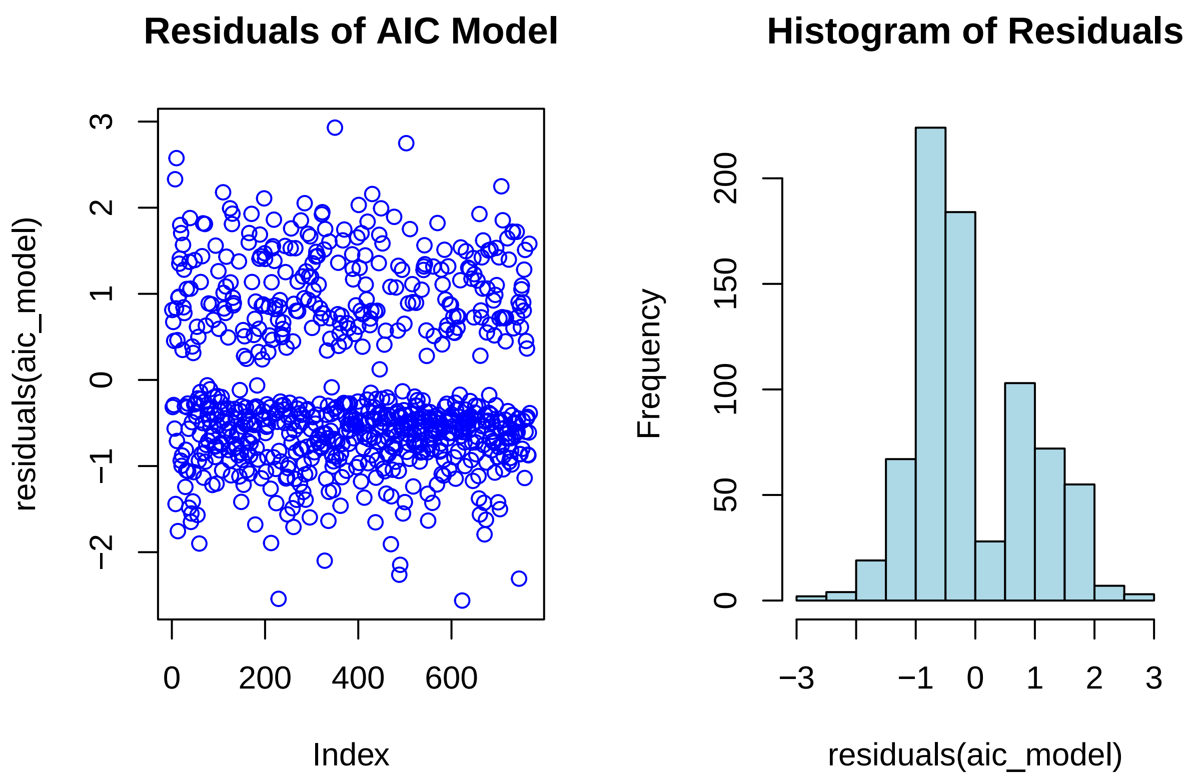
```
## Overall P-value: 9.651583e-54
```

```
## Pseudo R²: 0.2718097
```

Chunk

```
##
## Call:
## glm(formula = diabetes ~ pregnant + glucose + pressure + insulin +
##     mass + pedigree + age, family = binomial(link = "logit"),
##     data = PimaIndiansDiabetes)
##
## Coefficients:
##              Estimate Std. Error z value Pr(>|z|)
## (Intercept) -8.4051362  0.7167033 -11.727  < 2e-16 ***
```

```
## pregnant     0.1231724  0.0320688   3.841 0.000123 ***
## glucose      0.0351123  0.0036625   9.587  < 2e-16 ***
## pressure    -0.0132136  0.0051537  -2.564 0.010350 *
## insulin     -0.0011570  0.0008142  -1.421 0.155275
## mass         0.0900886  0.0144619   6.229 4.68e-10 ***
## pedigree     0.9475954  0.2980063   3.180 0.001474 **
## age          0.0147888  0.0092897   1.592 0.111393
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for binomial family taken to be 1)
##
##     Null deviance: 993.48  on 767  degrees of freedom
## Residual deviance: 723.45  on 760  degrees of freedom
## AIC: 739.45
##
## Number of Fisher Scoring iterations: 5
```



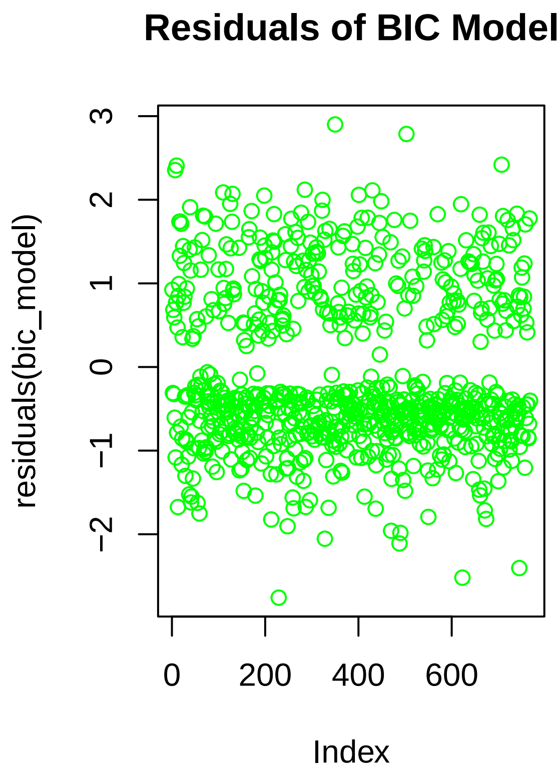
Chunk

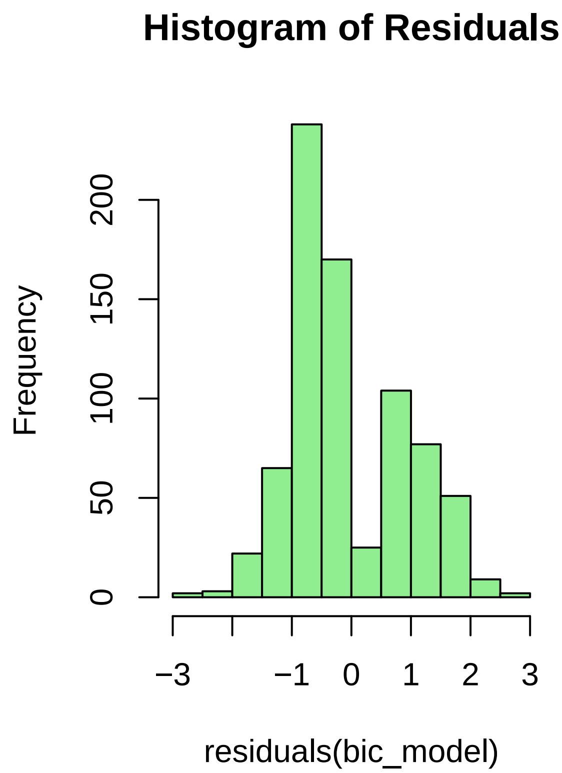
```
## Start:  AIC=783.24
## diabetes ~ pregnant + glucose + pressure + triceps + insulin +
##     mass + pedigree + age
##
##            Df Deviance    AIC
```

```
## - triceps   1   723.45 776.60
## - insulin   1   725.19 778.34
## - age       1   725.97 779.12
## - pressure  1   729.99 783.14
## <none>          723.45 783.24
## - pedigree  1   733.78 786.94
## - pregnant  1   738.68 791.83
## - mass      1   764.22 817.38
## - glucose   1   838.37 891.52
##
## Step:  AIC=776.6
## diabetes ~ pregnant + glucose + pressure + insulin + mass + pedigree +
##     age
##
##            Df Deviance    AIC
## - insulin   1   725.46 771.97
## - age       1   725.97 772.48
## <none>          723.45 776.60
## - pressure  1   730.13 776.64
## - pedigree  1   733.92 780.42
## + triceps   1   723.45 783.24
## - pregnant  1   738.69 785.20
## - mass      1   768.77 815.27
## - glucose   1   840.87 887.38
##
## Step:  AIC=771.97
## diabetes ~ pregnant + glucose + pressure + mass + pedigree +
##     age
##
##            Df Deviance    AIC
## - age       1   728.56 768.42
## <none>          725.46 771.97
## - pressure  1   732.51 772.37
## - pedigree  1   734.99 774.85
## + insulin   1   723.45 776.60
## + triceps   1   725.19 778.34
## - pregnant  1   741.27 781.13
## - mass      1   769.24 809.10
## - glucose   1   845.76 885.62
##
## Step:  AIC=768.42
## diabetes ~ pregnant + glucose + pressure + mass + pedigree
##
##            Df Deviance    AIC
## - pressure  1   734.31 767.52
## <none>          728.56 768.42
## - pedigree  1   738.43 771.65
## + age       1   725.46 771.97
## + insulin   1   725.97 772.48
## + triceps   1   728.00 774.51
## - pregnant  1   760.56 793.78
## - mass      1   770.21 803.43
## - glucose   1   862.96 896.18
##
```

```
## Step:  AIC=767.52
## diabetes ~ pregnant + glucose + mass + pedigree
##
##            Df Deviance    AIC
## <none>          734.31 767.52
## + pressure  1   728.56 768.42
## - pedigree  1   744.12 770.70
## + insulin   1   731.51 771.37
## + age       1   732.51 772.37
## + triceps   1   733.06 772.92
## - pregnant  1   762.87 789.45
## - mass      1   771.27 797.85
## - glucose   1   864.84 891.41
```

```
##
## Call:
## glm(formula = diabetes ~ pregnant + glucose + mass + pedigree,
##     family = binomial(link = "logit"), data = PimaIndiansDiabetes)
##
## Coefficients:
##              Estimate Std. Error z value Pr(>|z|)
## (Intercept) -8.415851   0.656908 -12.811  < 2e-16 ***
## pregnant     0.141926   0.027105   5.236 1.64e-07 ***
## glucose      0.033826   0.003345  10.112  < 2e-16 ***
## mass         0.078097   0.013771   5.671 1.42e-08 ***
## pedigree     0.901294   0.291696   3.090    0.002 **
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for binomial family taken to be 1)
##
##     Null deviance: 993.48  on 767  degrees of freedom
## Residual deviance: 734.31  on 763  degrees of freedom
## AIC: 744.31
##
## Number of Fisher Scoring iterations: 5
```





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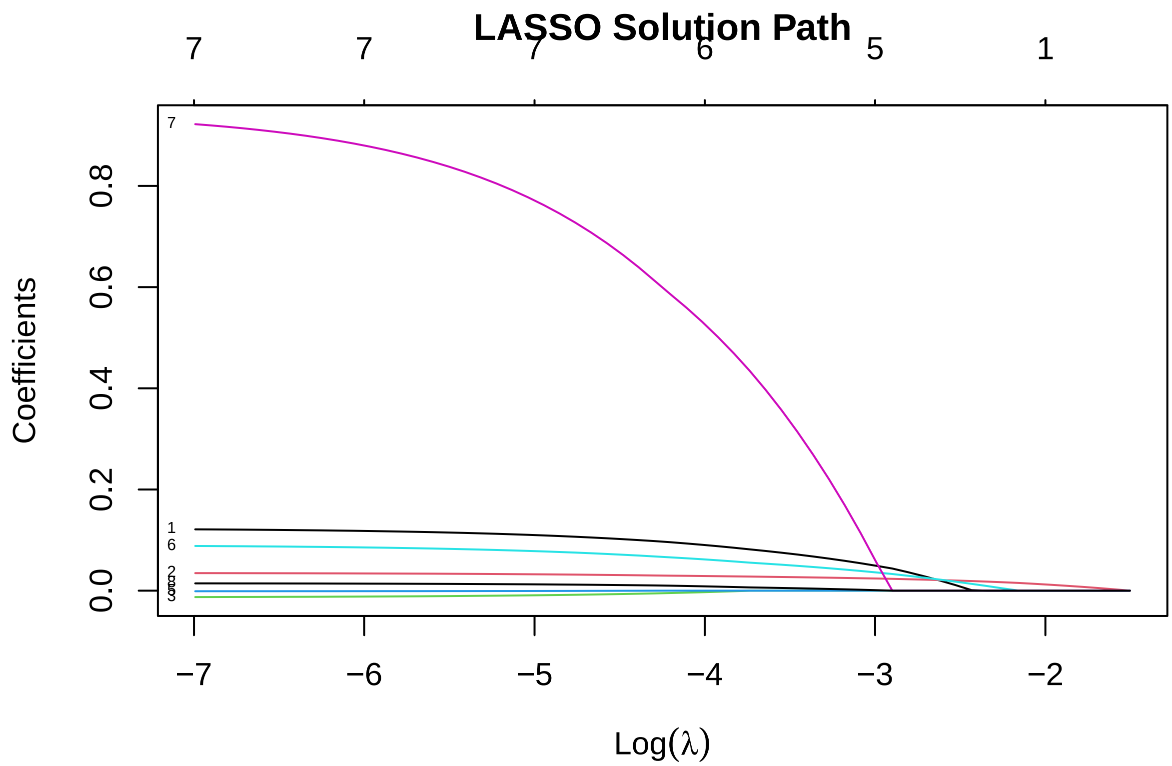
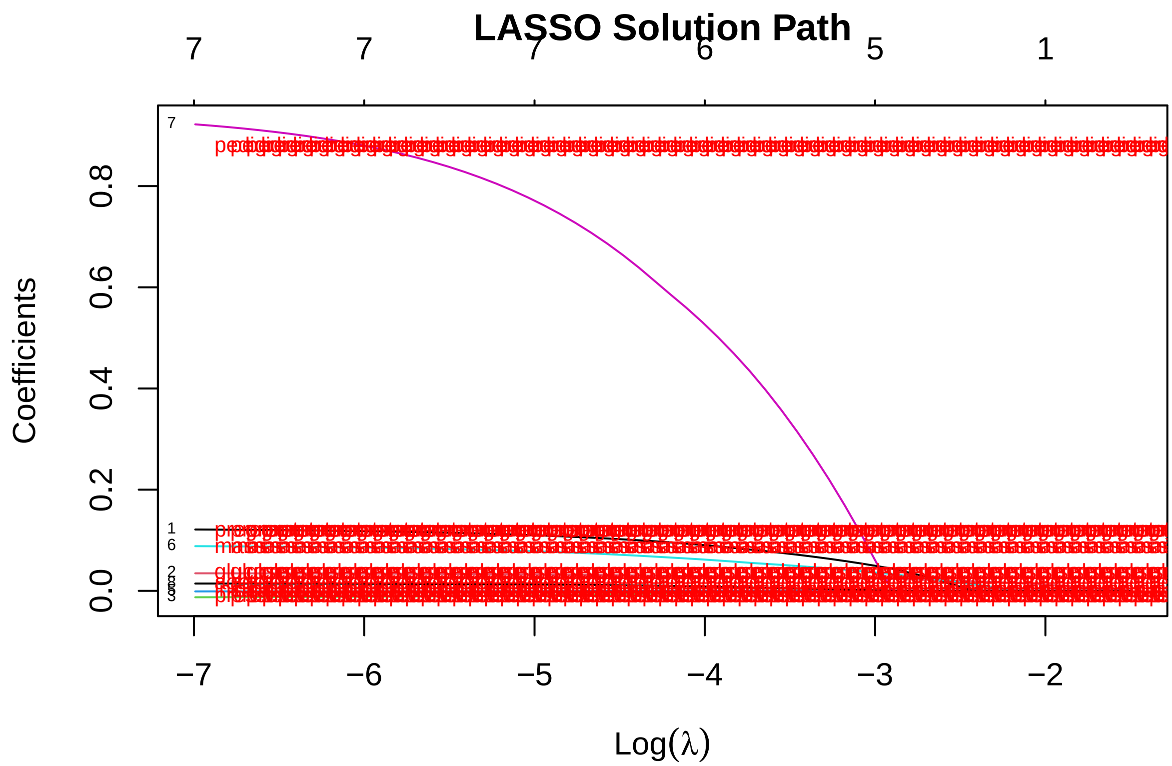
```
## Morgan-Tatar search since family is non-gaussian.

## BIC
## BICq equivalent for q in (0.169717182537119, 0.610343314895634)
## Best Model:
##                Estimate  Std. Error    z value     Pr(>|z|)
## (Intercept) -8.41585098 0.656907771 -12.811313 1.417163e-37
## pregnant     0.14192631 0.027105325   5.236104 1.640012e-07
## glucose      0.03382636 0.003345272  10.111690 4.903090e-24
## mass         0.07809694 0.013770941   5.671140 1.418501e-08
## pedigree     0.90129355 0.291696408   3.089834 2.002682e-03
```

Chunk



Chunk

```
## Optimal Lambda from Cross-Validation: 0.002556964

## LASSO Coefficients at Optimal Lambda:

## 9 x 1 sparse Matrix of class "dgCMatrix"
##                        s0
## (Intercept) -8.155802113
## pregnant     0.117910137
## glucose      0.034025961
## pressure    -0.011602618
## triceps      .
## insulin     -0.000935285
## mass         0.085348814
## pedigree     0.877808006
## age          0.013896219
```



Chunk

```
##
## Call:
## glm(formula = diabetes ~ pregnant + glucose + mass + pedigree,
##     family = binomial, data = data)
##
## Coefficients:
##              Estimate Std. Error z value Pr(>|z|)
## (Intercept) -8.415851   0.656908 -12.811  < 2e-16 ***
## pregnant     0.141926   0.027105   5.236 1.64e-07 ***
## glucose      0.033826   0.003345  10.112  < 2e-16 ***
## mass         0.078097   0.013771   5.671 1.42e-08 ***
## pedigree     0.901294   0.291696   3.090    0.002 **
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for binomial family taken to be 1)
##
##     Null deviance: 993.48  on 767  degrees of freedom
## Residual deviance: 734.31  on 763  degrees of freedom
## AIC: 744.31
##
## Number of Fisher Scoring iterations: 5
```

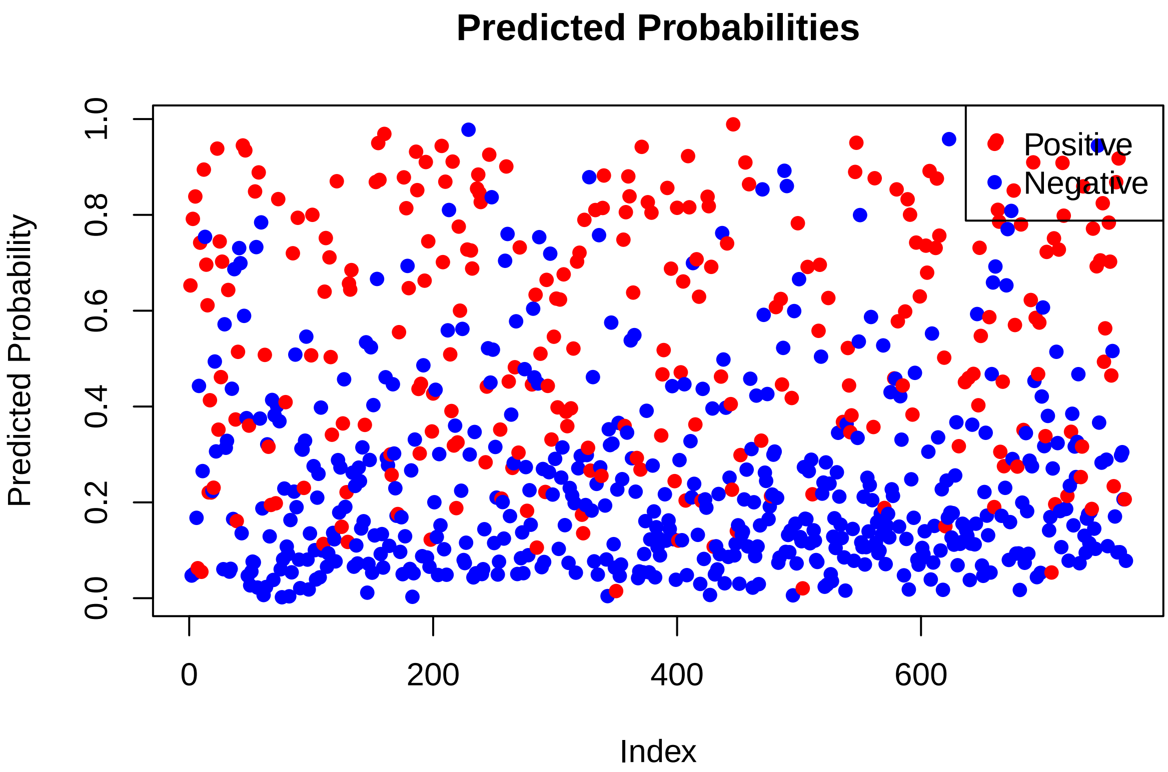
```
## Relative Change in Odds (in %):

## (Intercept)    pregnant     glucose        mass    pedigree
##  -99.977867   15.249172    3.440497    8.122747  146.278680

## For glucose: A one-unit increase leads to a 3.44 % change in the odds of diabetes.

## For pregnant: A one-unit increase leads to a 15.25 % change in the odds of diabetes.
```

Logistic Regression beyond the interpretation: Prediction and Performance assessment

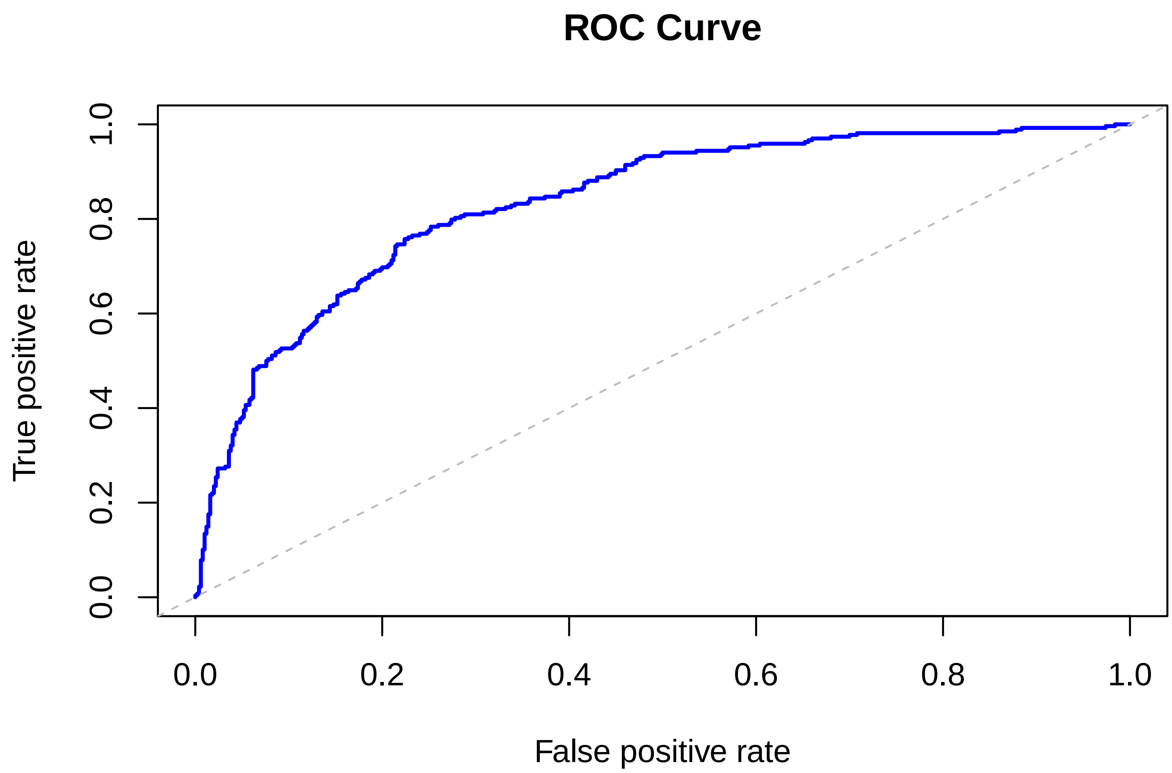
```
## Variables included in the BIC model:

## [1] "(Intercept)" "pregnant"    "glucose"     "mass"        "pedigree"
```

Chunk

```
##
## Call:
## glm(formula = diabetes ~ pregnant + glucose + mass + pedigree,
##     family = binomial, data = PimaIndiansDiabetes)
##
## Coefficients:
##              Estimate Std. Error z value Pr(>|z|)
## (Intercept) -8.415851   0.656908 -12.811  < 2e-16 ***
## pregnant     0.141926   0.027105   5.236 1.64e-07 ***
## glucose      0.033826   0.003345  10.112  < 2e-16 ***
## mass         0.078097   0.013771   5.671 1.42e-08 ***
## pedigree     0.901294   0.291696   3.090    0.002 **
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for binomial family taken to be 1)
##
##     Null deviance: 993.48  on 767  degrees of freedom
## Residual deviance: 734.31  on 763  degrees of freedom
## AIC: 744.31
##
## Number of Fisher Scoring iterations: 5

## Confusion Matrix:

##      Predicted
## True  neg pos
##   neg 442  58
##   pos 117 151
```



Predicted Probabilities
Positive
Negative
Predicted Probability
0.0
0.2
0.4
0.6
0.8
1.0
0
200
400
600
Index




```
## Area Under the Curve (AUC): 0.8345
```

```
##                       Metric     Value
## 1                   Accuracy 0.7721354
## 2                 Error Rate 0.2278646
## 3  True Positive Rate (TPR) 0.5634328
## 4 False Positive Rate (FPR) 0.1160000
## 5                  Precision 0.7224880
## 6                     Recall 0.5634328
## 7                    F-Score 0.6331237
```

```
## Model Evaluation Metrics:
```


```
##                       Metric     Value
## 1                   Accuracy 0.7721354
## 2                 Error Rate 0.2278646
## 3  True Positive Rate (TPR) 0.5634328
## 4 False Positive Rate (FPR) 0.1160000
## 5                  Precision 0.7224880
## 6                     Recall 0.5634328
## 7                    F-Score 0.6331237
```

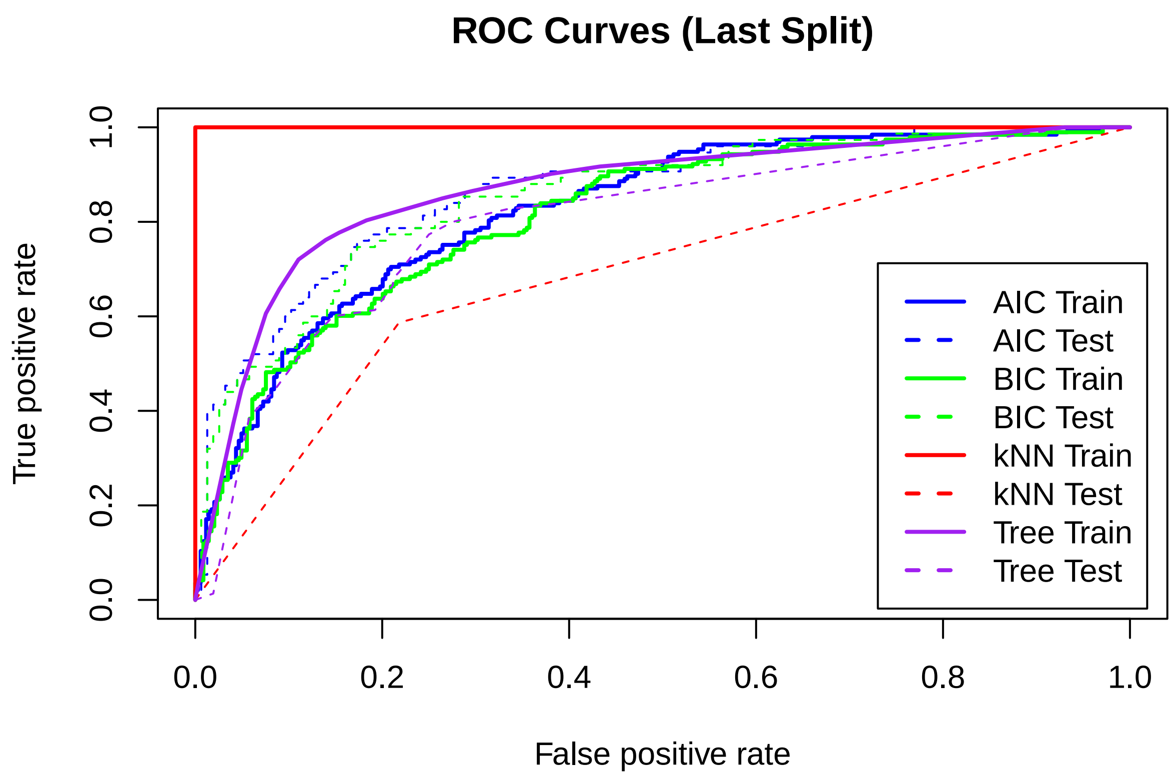
Predictive Analytics Full Scale

Chunk

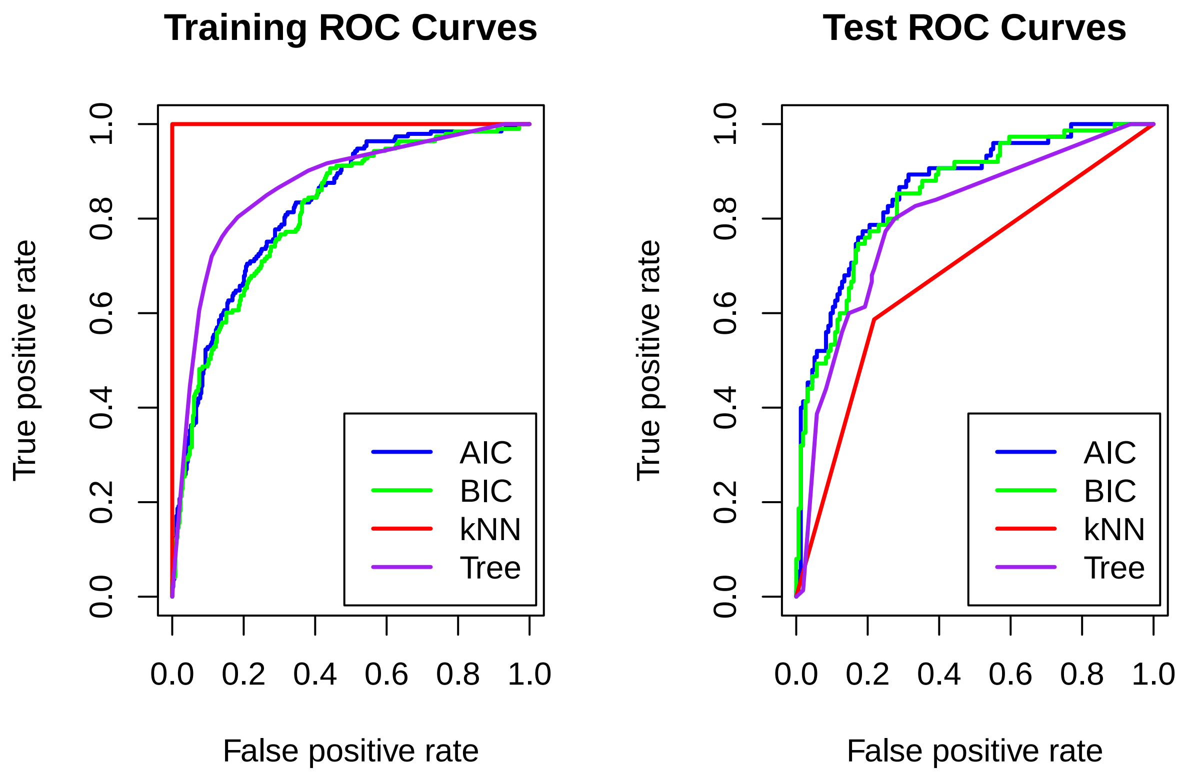
```
##
## Attaching package: 'dplyr'

## The following object is masked from 'package:car':
##
##     recode

## The following object is masked from 'package:MASS':
##
##     select

## The following objects are masked from 'package:stats':
##
##     filter, lag

## The following objects are masked from 'package:base':
##
##     intersect, setdiff, setequal, union
```

Subchunk

SubChunk



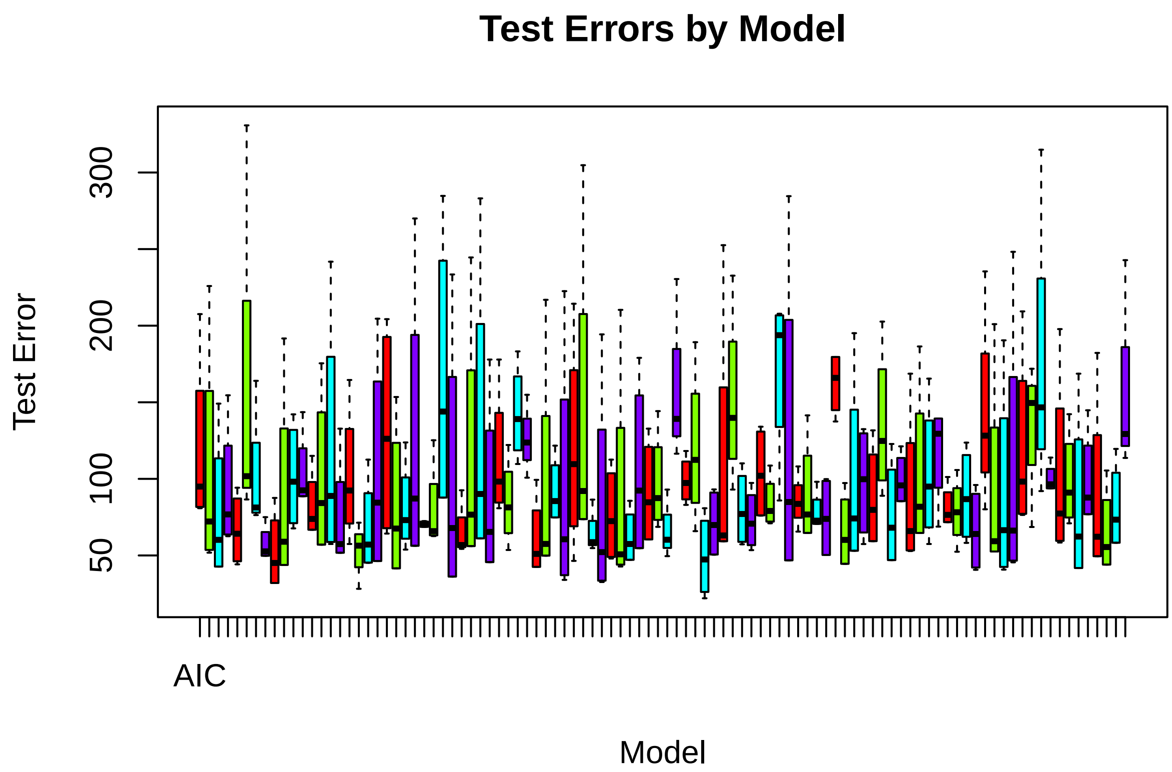
Chunk



Get the errors

```
## Completed 20 iterations
## Completed 40 iterations
## Completed 60 iterations
## Completed 80 iterations
## Completed 100 iterations

##
## Average Test Errors:

##        AIC       BIC       kNN      Tree
##   73.77842  67.81609 159.37444  93.95038

##
## Standard Deviations:

##      AIC      BIC      kNN     Tree
## 34.60298 30.98783 61.89843 30.98091
```

Prepare and boxplot the errors



Getting some F-scores

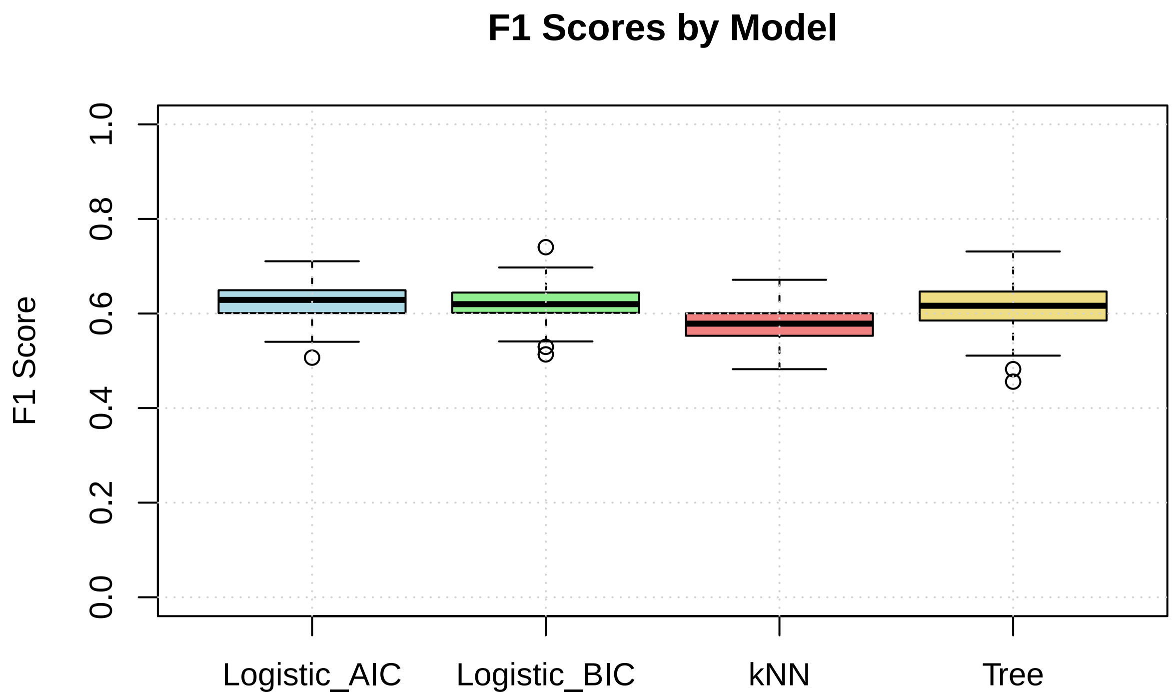
```
## Completed 20 F-score iterations
## Completed 40 F-score iterations
## Completed 60 F-score iterations
## Completed 80 F-score iterations
## Completed 100 F-score iterations
```

```
##
## Average F1 Scores (higher is better):
```

```
## Logistic_AIC Logistic_BIC          kNN         Tree
##    0.6259080    0.6195399    0.5770405    0.6127667
```

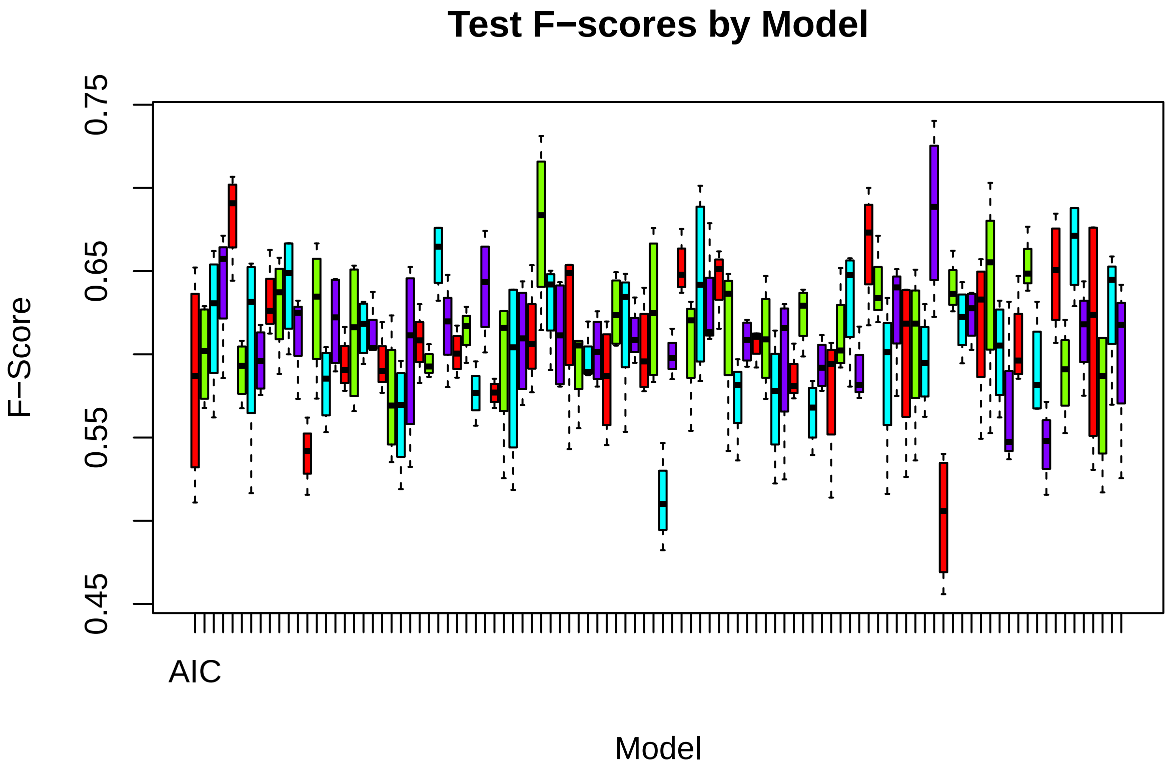
```
##
## Standard Deviations:
```

```
## Logistic_AIC Logistic_BIC          kNN         Tree
##   0.03721321   0.03636298   0.03609795   0.04520490
```

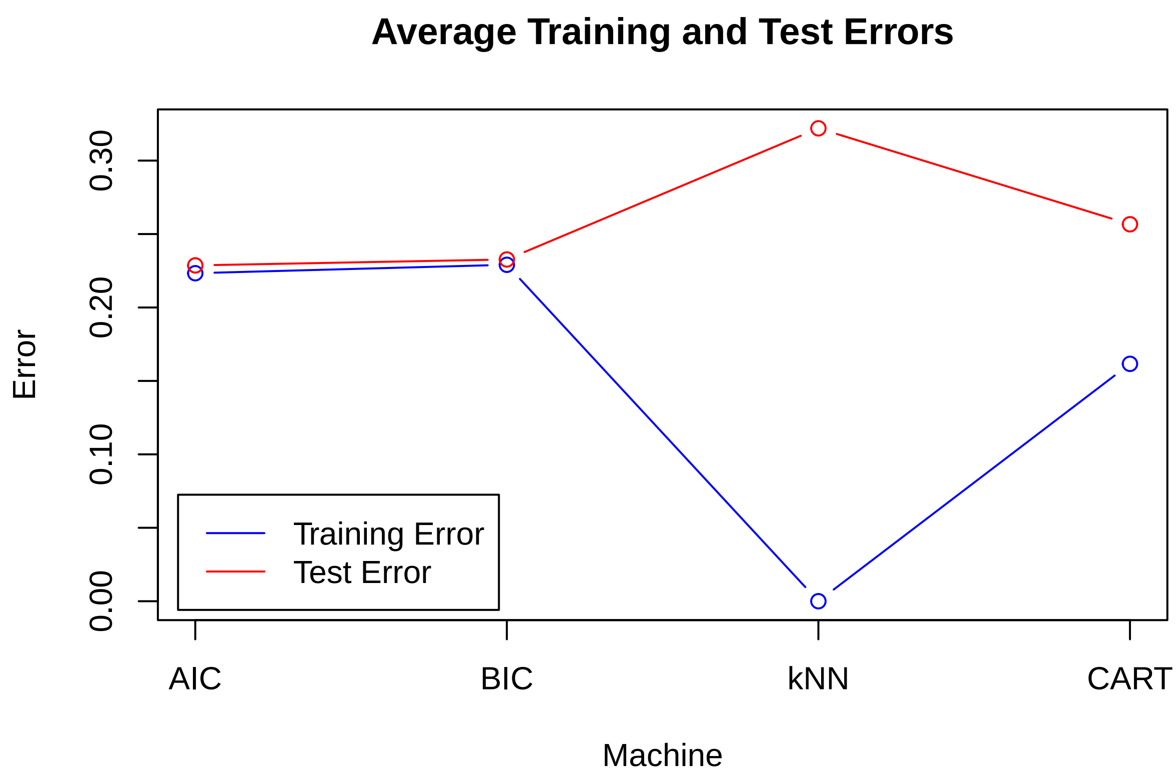


F1 Scores by Model
F1 Score
0.0
0.2
0.4
0.6
0.8
1.0
Logistic_AIC
Logistic_BIC
kNN
Tree

Showing the F-scores



Subchunk



Subchunk

Table 1: Average Training and Test Errors for Each Model

	Model	Avg Training Error	Avg Test Error
AIC	Logistic Regression (AIC)	0.2233	0.2287
BIC	Logistic Regression (BIC)	0.2291	0.2327
kNN	k-Nearest Neighbors	0.0000	0.3220
CART	Classification Tree	0.1616	0.2567

We are always Happy and Joyful! Thank you!